

Chau Dao

University of California, Los Angeles and Bhaumik Institute for Theoretical Physics

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Education

Ph.D. in Physics, specializing in Condensed Matter Theory **June 2026**

University of California, Los Angeles

Advisor: Yaroslav Tserkovnyak

Thesis: *Dynamics of Topological Textures*

M.S. in Physics **December 2020**

University of California, Los Angeles

B.S. in Physics, Computer Science, and Mathematics **May 2019**

Cornell University

Research Experience

Condensed Matter Theory and Quantum Information **University of Basel**
Supervised by Prof. Daniel Loss and Prof. Jelena Klinovaja *August 2026 –*

Condensed Matter Theory **UCLA**
Supervised by Prof. Yaroslav Tserkovnyak *September 2019 – August 2026*

Studied order parameter dynamics in a diverse range of physical systems, focused on exploring how topological order parameter textures can be induced and controlled in innovative ways. Extended theoretical techniques from spintronics to the liquid crystal community to predict a novel coupling between nematic and ionic dynamics in nematic electrolytes, opening up the field of nematronics. Studied the intertwining of spin degrees of freedom and the superconducting phase in ferromagnetic spin-triplet superconductors to devise a spin-triplet SQUID. Investigated the interplay of geometry and topological spin textures to study how system geometry can be employed to drive vortex transport on curved magnetic membranes.

Computational Condensed Matter Theory **University of Virginia**
Supervised by Prof. Gia-Wei Chern *June 2017 – May 2019*

Investigated the excitable network that controls the dynamics of membrane morphology of cells to understand how the cells respond to external stimulus. Studied dynamics of reaction-diffusion equation on a spherical surface by applying the FitzHugh-Nagumo model on a sphere. Studied the correlation between spiral waves on the spherical surface and spatiotemporal structures formed due to active noise. Developed a stochastic Crank-Nicolson numerical method to simulate the nonlinear stochastic system using Matlab.

Experimental High Energy Physics **UCLA**
Supervised by Prof. Jay Hauser *July 2019 – September 2019*

Analyzed potential firmware improvements that could be implemented for the CMS Endcap Muon detector. Using C++ and ROOT, independently developed a comprehensive codebase to emulate Anode Local Charged Track (ALCT) formation. Employed emulator to characterize the efficiency of current detection algorithms.

Experimental Biophysics **Cornell University**
Supervised by Prof. Robert Thorne *June 2017 – May 2019*

Studied the critical cooling rates and critical cooling concentrations of ternary solutions when submerged in liquid nitrogen or exposed to a shuttered gas stream. Performed high-precision experiments to investigate the effects of rapidly freezing protein solutions.

Selected Publications

Topological Hydrodynamics in Ferromagnetic Superconductors

C. Dao, E. Kleinherbers, B. Brekke, and Y. Tserkovnyak, *Phys. Rev. B* **113**, 174515 (2026)

Topological Transport of Vorticity on Curved Magnetic Membranes

C. Dao, J. Zou, E. Kleinherbers, and Y. Tserkovnyak, *Phys. Rev. B Lett.* **111**, L100401 (2025)

Nematronics: Reciprocal Coupling between Ionic Currents and Nematic Dynamics

C. Dao, J. C. Everts, M. Ravník, and Y. Tserkovnyak, *Phys. Rev. Lett.* **130**, 168102 (2023)

Nonabelian Topological Hydrodynamics

C. Dao, E. Kleinherbers, S. Kelly, S. Zhang, and Y. Tserkovnyak, *forthcoming*

Active Noise-Triggered Transient Localized Patterns in Reaction-Diffusion Systems

C. Dao, J.-M. Yang, C.-H. Huang, and G.-W. Chern, *forthcoming*

Topological Fate of Magnetic Instabilities

P. Connick, C. Dao, E. Kleinherbers, Y. Tserkovnyak, *forthcoming*

Topological Hydrodynamics in Condensed Matter Systems

C. Dao, E. Kleinherbers, J. Zou, Y. Tserkovnyak, *in preparation*

Advising and Academic Service

► Mentored and collaborated with graduate students as a coauthor on research projects, leading to forthcoming joint publications.

► Served as a reviewer for *Physical Review B* and *Physical Review Letters*.

Awards and Fellowships

Dissertation Year Fellowship, graduate student fellowship valued at \$20,000.

Invited Talks

Nonabelian Topological Hydrodynamics

April 2026

Mani-Fest 2026; Conference at University of California, Los Angeles

Intertwining Topology and Geometry: Topological Hydrodynamics on Curved Magnetic Membranes

March 2024

Seminar at University of Virginia

Contributed Talks and Poster Sessions

Nonabelian Topological Hydrodynamics

March 2026, Oral presentation

APS March Meeting 2026; Denver, Colorado; March 12-16, 2026

Intertwining Topology and Geometry: Topological Hydrodynamics on Curved Magnetic Membranes

August 2024, Oral Presentation

5th International Symposium on Advanced Magnetic Materials and Applications (ISAMMA); Dong Hoi City, Quang Binh Province, Vietnam; August 4-7, 2024

Topological Transport of Vorticity on Curved Magnetic Membranes

May 2024, Poster

Spin Caloritronics XIII; Beijing, China; May 19-24, 2024

Topological Transport of Vorticity on Curved Magnetic Membranes

March 2024, Oral Presentation

APS March Meeting 2024; Minneapolis, Minnesota; March 3-8, 2024

Nematronics: Reciprocal Coupling between Ionic Currents and Nematic Dynamics

March 2023, Oral Presentation

APS March Meeting 2023; Las Vegas, Nevada; March 5-10, 2023

Teaching Experience

► Taught **10 different courses** as a **Graduate Teaching Assistant** at **UCLA** for **20 academic quarters**, from **June 2019 to December 2025**. Courses ranged from introductory physics (classical mechanics, electromagnetism, waves and thermodynamics) to graduate-level electives, and included theoretical courses as well as laboratory courses. Managed courses with over 200 students as the sole teaching assistant. Selected courses including course descriptions and teaching duties listed below:

PHYSICS 221C – Graduate Quantum Mechanics

UCLA

Graduate Teaching Assistant

Spring 2023

Wrote exams and homeworks, prepared and led discussion sections. Assisted professor in designing the course structure, substituted for the professor as the main lecturer. Elective course for advanced graduate students covering quantum theory of radiation, introduction to relativistic quantum mechanics, second quantization, and elements of many-body theory.

PHYSICS 18L – Modern Physics Laboratory

UCLA

Graduate Teaching Assistant

Fall 2021, Fall 2020

Designed laboratory experiment with professors, graded lab reports, demonstrated experiments. Experimental laboratory course for advanced undergraduate students focused on radioactivity, scattering, Planck constant, superconductivity, superfluidity.

PHYSICS 180F – Elementary Particle Laboratory

UCLA

Graduate Teaching Assistant

Winter 2021

Designed laboratory experiment with professors, graded lab reports, demonstrated experiments. Experimental laboratory course for advanced undergraduate students, with a focus on an overview of standard model, interaction of particles with matter, basic particle detection techniques, introductory probability and statistics, and analysis with Python and ROOT.

► **Undergraduate teaching assistant** and grader at **Cornell University** for upper-level courses in the Department of Physics and Department of Computer Science from **2017 to 2019**. Courses with descriptions and teaching duties listed below:

CS 4850: Mathematics for the Information Age

Cornell University

Teaching Assistant

Spring 2019

Graded homeworks, exams, and held weekly office hours. Joint course for graduate students and advanced undergraduates on the mathematics underpinning machine learning and data science: high-dimensional statistics, principal component analysis, Markov chains, VC dimension, random graphs, and low-rank matrix factorizations.

AEP 4210: Mathematical Physics I

Cornell University

Teaching Assistant

Fall 2017

Wrote homework solutions, graded homework assignments and exams for an upper undergraduate/master's level course on mathematical methods used for physics. The course provides students with the requisite mathematical background needed to take higher level physics courses.

Skills

Computational skills and programming languages

- ▶ Numerical Analysis, Multivariate Data Analysis, Classical Monte Carlo Methods
- ▶ C/C++, mumax, Python, Java, Bash, AutoCAD, OCaml, Mathematica, Matlab, ROOT, L^AT_EX

Languages | Vietnamese (native) ◦ English (native) ◦ French (basic)